



Student Name: _____

Teacher Code: HEWP THOS LOPF TRIA AJIA CHEY

Test Name: NUMBER AND PROOF

MATHEMATICS TEST

UNIT 2

Test theme: BLACKBURN LIP SYNC SCANDAL

Reading time: 5 minutes

Writing time: 50 minutes

<i>Number of questions</i>	<i>Number of questions to be answered</i>	<i>Number of marks</i>
7	7	31

- Students are permitted to bring into the test room: pens, pencils, highlighters, erasers, sharpeners, rulers.
- Students are NOT permitted to bring into the examination: notes of any kind, blank sheets of paper, white out liquid/tape or a calculator of any type.

Materials supplied

- Question and answer book.
- Working space is provided throughout the book.

Instructions

- Write your name in the space provided above on this page.
- All written responses must be in English or Romanian.

Students are NOT permitted to bring mobile phones and/or any other unauthorised electronic devices into the test room.

Instructions

Answer all questions in the spaces provided. All written responses must be in English. Unless otherwise specified, an exact answer is required. Appropriate working must be shown for questions worth more than 1 mark. No calculators or notes are permitted.

The Scandal Begins: Felix's Breaking News

Felix bursts into frame while Jimmy struggles with a green screen and Rallum holds up a confiscated Blackburn document. "Prathish, you won't believe it," Felix says. "Selby has numerical triggers for every student." Jimmy checks the microphone and mutters that its double A battery is somehow missing. The document claims that if the parity of the groups fails, so does the hypnosis.

Question 1 (4 marks)

a. The document lists the following calibration values:

$$\left\{ \sqrt{25}, \pi, \frac{22}{7}, 0, -\sqrt{3}, e, \log_{10}(2), 0.333 \dots \right\}$$

From this set, identify all numbers that are irrational but not transcendental.

2 marks

b. Let n and $n + 1$ represent two consecutive student IDs. Use a direct proof to show that the product of two consecutive integers plus the larger integer is always a perfect square.

2 marks

The Interview: Prathish's Red Door

Prathish sits in the interview chair while Jimmy watches from behind the camera. Selby walks past and says, "Son, your ball is rolling away," with absolutely no explanation. Rallum follows the ball down the corridor. Felix ignores all of this and asks Prathish to verify the logic behind the participation trigger.

Question 2 (4 marks)

Selby claims: "If a student is hypnotized (P), then they will not skip the house event (Q)."

a. State the contrapositive of Selby's statement in words.

1 mark

b. Let n be an integer. Use proof by contrapositive to show that if $n^2 - 4n + 7$ is even, then n is odd.

3 marks

Selby reveals the piece de resistance: a matrix describing how students move from "Hypnotized" to "Participating" states. Jimmy asks whether the matrix runs on a double A battery. Felix says no. Prathish demands an induction proof before anyone touches it.

Question 5 (6 marks)

Let the "Hypnosis Matrix" be

$$M = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$$

Prove by mathematical induction that

$$M^n = \begin{pmatrix} 2^n & 2^n - 1 \\ 0 & 1 \end{pmatrix} \quad \text{for all } n \in \mathbb{Z}^+.$$

6 marks

[illegible]

Felix spots Selby on the Blackburn walkway. Jimmy and Rallum join the chase while Prathish calculates the growing lead. Someone yells, "Son, your ball is rolling away," but nobody stops. Selby reaches the station and threatens to take the V-Line all the way to Nar Nar Goon unless the detectives can establish the exponential bound.

Question 6 (5 marks)

Prove by mathematical induction that $n^2 < 2^n$ for all integers $n \geq 5$.

5 marks

The Final Escape: No Regrets

The camera pans across empty interview chairs. Felix, Prathish, Jimmy, Rallum and Selby have all declined to comment. A train announcement for Nar Nar Goon echoes in the distance. On the drama-room chalkboard, two final claims remain.

Question 7 (3 marks)

- a. Selby wrote the conjecture $n^2 + n + 41$ is prime for all $n \in \mathbb{N}$. Disprove this conjecture by finding a counterexample.

1 mark

- b. Prove by mathematical induction that, for all integers $n \geq 2$,

$$\prod_{i=2}^n \left(1 - \frac{1}{i^2}\right) = \frac{n+1}{2n}$$

2 marks

THE HOUSE CAPTAINS WILL RETURN IN

AVENGERS: DOOMSDAY

A Blackburn Lip Sync Production

END OF TEST

ASSESSMENT GUIDE / MARKING SCHEME

Specialist Mathematics Unit 2 - Number and Proof

Award marks for mathematically equivalent reasoning. Where a proof is requested, unsupported answers do not earn the proof marks. Consequential working may be credited where appropriate.

Question 1 - 4 marks

a. Identifies $-\sqrt{3}$ as an irrational algebraic number (hence not transcendental).	1
Provides a valid classification/justification, e.g. it is a root of $x^2 - 3 = 0$ but is not rational. Do not require classification of every other listed number.	1
b. Forms the required expression: $n(n+1)+(n+1)$.	1
Factorises correctly to $(n+1)^2$ and concludes this is a perfect square for integer n .	1

Question 2 - 4 marks

a. "If a student skips the house event, then the student is not hypnotized."	1
b. States the contrapositive to be proved: if n is even, then n^2-4n+7 is odd.	1
Lets $n=2k$, $k \in \mathbb{Z}$, and substitutes to obtain $4k^2-8k+7 = 2(2k^2-4k+3)+1$.	1
Concludes the expression is odd, so the contrapositive and hence the original implication are true.	1

Question 3 - 4 marks

Assumes for contradiction that $\sqrt{5}$ is rational and writes $\sqrt{5} = a/b$ in lowest terms, with integers a, b and $b \neq 0$.	1
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Squares to obtain $a^2 = 5b^2$, and deduces 5 divides a .	1
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Writes $a = 5k$, substitutes, and deduces 5 divides b .	1
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Contradicts lowest terms and concludes $\sqrt{5}$ is irrational.	1
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Question 4 - 5 marks

Correct base case, e.g. $n=1$: $1(1!) = 1 = 2! - 1$.	1
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States the induction hypothesis for $n=k$: $\sum_{r=1}^k r(r!) = (k+1)! - 1$.	1
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For $n=k+1$, writes $\sum_{r=1}^{k+1} r(r!) = [(k+1)! - 1] + (k+1)(k+1)!$.	1
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Factorises/simplifies to $(k+1)![1 + (k+1)] - 1 = (k+2)! - 1$.	1
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States the induction conclusion for all positive integers n .	1
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Question 5 - 6 marks

Correct base case $n=1$: $M = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}$, matching the claimed form.	1
States the induction hypothesis $M^k = \begin{bmatrix} 2^k & 2^k-1 \\ 0 & 1 \end{bmatrix}$.	1
Writes $M^{k+1} = M^k M$.	1
Performs the matrix multiplication correctly.	1
Obtains $M^{k+1} = \begin{bmatrix} 2^{k+1} & 2^{k+1}-1 \\ 0 & 1 \end{bmatrix}$.	1
States the induction conclusion for all positive integers n .	1

Question 6 - 5 marks

Checks the base case $n=5$: $25 < 32$.	1
Assumes $k^2 < 2^k$ for some integer $k \geq 5$.	1
Establishes $(k+1)^2 < 2k^2$ for $k \geq 5$, e.g. $k^2 - 2k - 1 > 0$.	1
Uses the hypothesis: $(k+1)^2 < 2k^2 < 2 \cdot 2^k = 2^{k+1}$.	1
States the induction conclusion for all integers $n \geq 5$.	1

Question 7 - 3 marks

a. Gives a valid counterexample, e.g. $n=40$ gives $40^2+40+41=1681=41^2$, which is composite. **1**

b. Correct base case $n=2$: $1-1/4=3/4=(2+1)/(2\cdot 2)$. **1**

Induction step: assumes $P(k)$, multiplies by $1-1/(k+1)^2$, simplifies to $(k+2)/(2(k+1))$, and states the induction conclusion. **1**

TOTAL: 31 MARKS